

(6.2)

Petr Yára

$$V = \left\{ \begin{pmatrix} v_1 \\ 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} v_2 \\ 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} \right\} \quad [f]_V = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

$\lambda_{11}, a_{12}, a_{21}, a_{22}; \lambda \in \mathbb{R}$

Kjistiíme nejprve $[f]_V$

$$a_{11}v_1 + a_{21}v_2 = \begin{pmatrix} 4 \\ 1 \\ 2 \\ 5 \end{pmatrix}$$

$$\rightarrow \begin{pmatrix} 2 & 1 & 4 & 2 \\ 1 & 0 & 1 & 2 \\ 0 & 1 & 2 & -2 \\ 1 & 2 & 5 & -2 \end{pmatrix}$$

$$\rightarrow \begin{pmatrix} 2 & 1 & 4 & 2 \\ 1 & 0 & 1 & 2 \\ 0 & 1 & 2 & -2 \\ 1 & 2 & 5 & -2 \end{pmatrix}$$

2. a 3. dávají výsledek 1; 4. kontrola

$$a_{11}=1 \quad a_{12}=2 \quad (1) \quad 2+2=4 \checkmark; \quad 4-2=2 \checkmark$$

$$a_{21}=2 \quad a_{22}=-2 \quad (4) \quad 1+4=5 \checkmark; \quad 2-4=-2 \checkmark$$

$$\text{tedy } [f]_V = \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix}$$

$$\det([f]_V - \lambda I) = (1-\lambda)(-2-\lambda) - 4 =$$

$$\text{tedy } \lambda_1 = -3; \lambda_2 = 2$$

$$= \lambda^2 + \lambda - 6 = (\lambda + 3)(\lambda - 2)$$

$$\lambda_1: \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -3x \\ -3y \end{pmatrix}$$

$$x + 2y = -3x$$

$$2x - 2y = -3y \Rightarrow y = -2x \quad \text{tedy } m_{\lambda_1} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$\lambda_2: \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} c \\ p \end{pmatrix} = \begin{pmatrix} 2c \\ 2p \end{pmatrix}$$

$$c + 2p = 2c \Rightarrow c = 2p$$

$$\text{tedy } m_{\lambda_2} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 1 & 5 \\ 1 & 0 & 1 \\ 0 & 1 & 3 \\ 1 & 2 & 7 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix}_V = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$(1) \quad 2+3=5 \checkmark$$

$$(4) \quad 1+6=7 \checkmark$$

podle věty pak dostáváme rozklad $[f^n]_V = R D R^{-1}$ kde R má sloupce vlastní vektory přísl. vlastním číslem u matice D která je diag.

$$R = \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}$$

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ -2 & 1 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 5 & 2 & 1 \end{array} \right) \sim \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & \frac{2}{5} & \frac{1}{5} \end{array} \right)$$

$$D^n = \begin{pmatrix} (-3)^n & 0 \\ 0 & 2^n \end{pmatrix}$$

$$\sim \left(\begin{array}{cc|cc} 1 & 0 & \frac{1}{5} & -\frac{2}{5} \\ 0 & 1 & \frac{2}{5} & \frac{1}{5} \end{array} \right) \Rightarrow P^{-1} = \frac{1}{5} \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}$$

$$\left[\begin{array}{c} 1 \\ 1 \\ 3 \\ 7 \end{array} \right]_V \cdot \left[\begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} \right]_V =$$

$$\begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} (-3)^n & 0 \\ 0 & 2^n \end{pmatrix} \cdot \frac{1}{5} \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \end{pmatrix} =$$

$$= \frac{1}{5} \begin{pmatrix} (-3)^n & 2^{n+1} \\ -2 \cdot (-3)^n & 2^n \end{pmatrix} \cdot \begin{pmatrix} -5 \\ 5 \end{pmatrix} = \frac{5}{5} \begin{pmatrix} (-3)^n & 2^{n+1} \\ (-3)^n \cdot (-2) & 2^n \end{pmatrix} \begin{pmatrix} -1 \\ 1 \end{pmatrix} =$$

$$= \begin{pmatrix} -(-3)^n + 2^{n+1} \\ (-3)^n \cdot 2 + 2^n \end{pmatrix}$$

chceme to ale v kanonické bázi tedy

$$[id]_K^V = \begin{pmatrix} 2 & 1 \\ 1 & 0 \\ 0 & 1 \\ 1 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 1 \\ 1 & 0 \\ 0 & 1 \\ 1 & 2 \end{pmatrix}$$

$$\cdot \begin{pmatrix} -(-3)^n + 2^{n+1} \\ (-3)^n \cdot 2 + 2^n \end{pmatrix} =$$

$$= \begin{pmatrix} -2(-3)^n + 2^{n+2} + 2(-3)^n + 2^n \\ -(-3)^n + 2^{n+1} \\ (-3)^n \cdot 2 + 2^n \\ -(-3)^n + 2^{n+1} + 4(-3)^n + 2^{n+1} \end{pmatrix}$$

$$= \begin{pmatrix} 5 \cdot 2^n \\ -(-3)^n + 2^{n+1} \\ 2(-3)^n + 2^n \\ 3(-3)^n + 2^{n+2} \end{pmatrix} = \begin{array}{c} \cancel{1}^n \begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} \end{array}$$